

# 3.1 Programming & Language Types

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 01 of 13

## 3.1 Programming & Language Types

### LEARN IT

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own. They need exact, step-by-step **instructions**.

- A set of instructions given to a computer is called a **program**.
- The language used to write these instructions is called a **programming language**.
- Writing programs is called **programming**, and the person who writes them is a **programmer**.

Think of a recipe. A cook follows the recipe line by line: "boil water, add rice, wait 15 minutes". A computer follows a program in the same way, one instruction at a time, without guessing what you meant.

### WORKED EXAMPLE

**A tiny program.** Here is a complete Python program with one instruction. It tells the computer to show a message on the screen.

```
print("Hello")
```

OUTPUT  
Hello

The word `print` is the instruction; `"Hello"` is the data it works on.

### REMEMBER IT

**Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly.** If the recipe is wrong, the food is wrong. The cook does not fix it for you.

### COMMON MISTAKE

Students often think the computer "understands" what they want. It does not. If you type `Print("Hello")` with a capital P, Python stops with an error, because it only knows `print`. Be exact.

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔ اسے کام کرنے کے لیے ہدایات چاہئیں۔ ہدایات کے مجموعے کو پروگرام (Program) کہتے ہیں، اور جس زبان میں یہ ہدایات لکھی جاتی ہیں اسے پروگرامنگ لینگویج (Programming Language) کہتے ہیں۔

#### CHECK YOURSELF

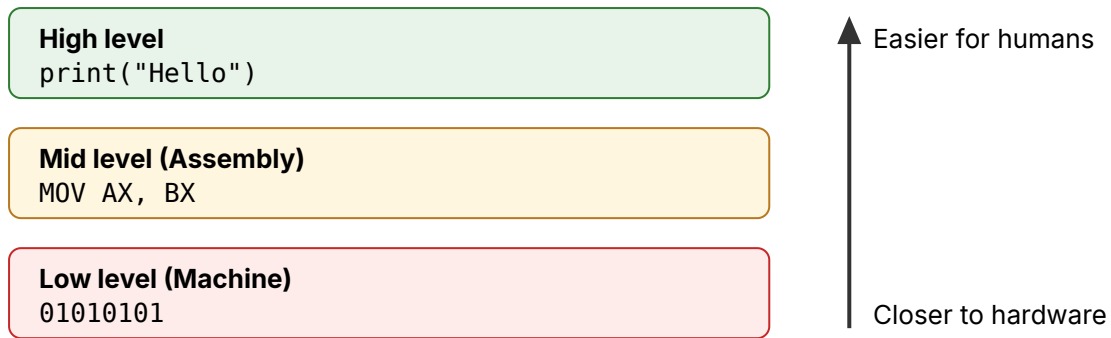
1. What is a program?
2. Why do we say computers are "dumb" machines?

Answers: 1. A set of instructions given to a computer to perform a task. 2. Because they cannot think or do anything without instructions; they only follow exactly what they are told.

#### LEARN IT

Programming languages are classified into **three main types**, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

| Type              | Description                                                                                                                                                                | Examples                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Low level</b>  | Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.                                                            | Machine language (binary) |
| <b>Mid level</b>  | Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called <b>mnemonics</b> (e.g. <code>MOV</code> , <code>ADD</code> ). | Assembly language         |
| <b>High level</b> | Designed to be easy for humans to read and write. Uses English-like words.                                                                                                 | C++, Java, Python         |



The three levels of programming languages, with a sample line from each.

## Common high-level languages

| Language | Key idea                         | Used for                                                                         |
|----------|----------------------------------|----------------------------------------------------------------------------------|
| C++      | Powerful and closer to hardware. | System software, operating systems, high-performance computer games.             |
| Java     | "Write once, run anywhere."      | Large enterprise applications, Android mobile apps, server programming.          |
| Python   | Simple, readable syntax.         | Data Science, Artificial Intelligence, Web Development, and learning to program. |

### WORKED EXAMPLE

**Same task, three levels.** Suppose you want to copy a value.

- Machine language: a pattern like `01010101` — you would need a chart to know what it means.
- Assembly: `MOV AX, BX` — "move the value in register BX into register AX". Shorter words, but you still need to know the hardware.
- Python: `a = b` — you simply say "a gets the value of b".

The higher the level, the less you need to know about the hardware and the faster you can write the program.

### REMEMBER IT

**"Low = Like the machine, High = Human."** Mid level sits in the Middle and uses Mnemonics (both start with M).

### COMMON MISTAKE

"Low level" does not mean "bad" or "weak". Low-level code runs very fast and controls hardware directly. It is just harder for humans to write.

### اردو میں سمجھیں

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الفاظ (Mnemonics) استعمال کرتی ہے، اور ہائی لیول (High Level) جیسے پائتھن، جاوا اور سی پلس پلس جو انگریزی جیسے الفاظ استعمال کرتی ہیں اور انسانوں کے لیے آسان ہیں۔

### CHECK YOURSELF

1. Which level is `MOV AX, BX` ?
2. Which high-level language is famous for "write once, run anywhere"?
3. Give one use of C++.

Answers: 1. Mid level (assembly language). 2. Java. 3. Any one: system software, operating systems, high-performance games.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** A program is:

- a. A set of instructions
- b. A mouse
- c. A cable
- d. A pie

**MCQ 2.** High-level languages are:

- a. Closer to English (Python, C++)
- b. Only 0 and 1
- c. Only analog
- d. Only SQL

**MCQ 3.** Machine language is:

- a. Binary codes for the CPU
- b. Python
- c. Urdu
- d. HTML always

**MCQ 4.** C++ is typically:

- a. High-level, closer to hardware than Python
- b. A DBMS
- c. A chart
- d. An OSI layer

**MCQ 5.** Python is popular in XI because:

- a. Simple syntax, huge libraries
- b. It is binary only
- c. It cannot print
- d. It is analog

**MCQ 6.** The translator of Python is a:

- a. Interpreter (also compiles to bytecode)
- b. Router
- c. Switch
- d. K-map

## Extra short questions

1. Define program and programming language.
2. Differentiate high-level and low-level.
3. Name three high-level languages and one use each.
4. Why are computers "dumb" without programs?

## Extra long question

1. Compare machine, assembly and high-level languages in a table (readability, speed, portability, example). Link why this course uses Python.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Program: instructions for a computer. Language: the notation used to write them.
2. High-level: English-like, portable. Low-level: close to hardware, faster but harder.
3. Python data/scripts; C++ systems/games; Java apps — (accept similar).
4. They only follow instructions; they do not understand the problem.

**Long**

1. Machine 0/1 fastest least portable; assembly mnemonics; high-level portable. Python: readable, batteries included, matches the Sindh practicals.

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Next lecture: 3.2 Career Growth with Python